The conserved background of the first memory return
A strong leading limit for ordinary charge propagation

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Abstract

In the common-coupling complex double-scaled SYK model, we upgrade the energy regression of a summed force sandwich to an annealed L^2 approximation, uniformly on compact real-time intervals. Two tagged force pairs select two prescribed, uncrossed bubbles; the established q-Mehler conditional expectation then supplies their limit. The first memory return converges to an explicit conserved function. Exact energy pinching therefore reduces the dynamic memory field to its ordinary-propagation term in L^2. The remaining problem is the subleading fluctuation of that term.

1 Model, state and precise scope

Use standard CAR, $\hbar = 1$, string time, and the overlapping model

$$H_N = \sum_{I,J} J_{IJ} c_I^\dagger c_J, \qquad J = J^\dagger, \tag{1}$$

$$\mathbb{E} J_{IJ} J_{KL} = w_N \delta_{IL} \delta_{JK}, \qquad w_N = 4^p \frac{J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2. \tag{2}$$

Here I, J are increasing p-subsets and $c_J = (c_J^\dagger)^\dagger$; the complex-SYK framework follows [2]. Take even N, $\lambda_N = p^2/N \to \lambda > 0$ in a positive compact interval, with $J_s > 0$ fixed. All charge sectors share the same J. The state is the full normalized annealed trace at $\beta = \mu = 0$:

$$\|X\|_r^r = \mathbb{E}\, 2^{-N} \operatorname{Tr} |X|^r = \sum_m b_m \mathbb{E} \operatorname{tr}_m |X_m|^r, \qquad b_m = 2^{-N} \binom{N}{m}. \tag{3}$$

For $m > 0$ set $h = H_m$, $V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi$, $K = 1_N \otimes H_{m-1}$, $Q = 1 - VV^\dagger$, and

$$a = V^\dagger K V = (1 - p/m) h, \qquad R = QKV, \qquad D = QKQ, \tag{4}$$

$$G = R^\dagger R, \qquad M(s) = R^\dagger e^{iDs} R, \qquad \Lambda(s) = R^\dagger e^{iKs} R. \tag{5}$$

Let $\mathsf{P}_h X = \sum_e P_e X P_e$, retaining entire degenerate blocks, and define

$$\mathsf{F}_m(s) = (1 - \mathsf{P}_h) m e^{-ihs/2} M(s) e^{-ihs/2}, \quad X_{0,m}(s) = (1 - \mathsf{P}_h) m e^{-ihs/2} \Lambda(s) e^{-ihs/2}. \tag{6}$$

Inactive $m < p$ sectors are zero. We prove $\sup_{|s| \le S} \|\mathsf{F}(s) - X_0(s)\|_2 \to 0$ for every fixed S. This concerns the specified microscopic model; no finite-G bulk dynamics or full de Sitter dictionary is inferred.

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2 Two tagged bubbles give a strong polynomial limit

Put $F_i = [H_N, c_i]$, $\mathcal{A} = \sum_i F_i^\dagger F_i$, and

$$\mathcal{Q}_N(f) = \frac{2p}{N} \sum_i F_i^\dagger f(H_N) F_i. \tag{7}$$

The middle Hamiltonian automatically acts on the lower charge sector. Let $q = e^{-4\lambda}$, $\sigma^2 = J_s^2/\lambda$, and ν_q be the variance-σ^2 q-Gaussian law. Its monic Hermite polynomials satisfy $xW_n = W_{n+1} + \sigma^2[n]_q W_{n-1}$. The standard Mehler operator obeys $\mathsf{P}_q W_n = q^n W_n$ [1]. We claim, for every fixed polynomial f,

$$\|\mathcal{Q}_N(f) - 4J_s^2(\mathsf{P}_q f)(H_N)\|_2 \longrightarrow 0. \tag{8}$$

Expand $\mathbb{E}\tau\mathcal{Q}_N(f)^\dagger\mathcal{Q}_N(g)$ by Gaussian pairings. Each chord has independent uniform p-sets I_α, J_α; the covariance identifies only its two endpoints. A force tag must occur at both endpoints of its own force pair. If these endpoints are paired to different chords, the tag belongs to the intersection of two independent sets A, B. The two-tag count is then at most $p|A \cap B|$, instead of p^2 for two internal pairs.

This bound still applies when tags coincide, their constraints coincide, they share a chord, or a force endpoint pairs with a background H. Accidental equality of independent chord labels is included in their set sums; it is not an additional covariance identification. For fixed word length L, let Z sum all pairwise intersections of the independent sets. Local CAR trace weights, relative to the disjoint-site normalization, are bounded by $C_L e^{C_L Z}$. Hypergeometric factorial moments and Hölder give

$$\sup_N \mathbb{E}e^{tZ} < \infty, \qquad \sup_N \mathbb{E}\big[|A \cap B| e^{tZ}\big] < \infty \quad (t \text{ fixed}). \tag{9}$$

Thus every noninternal pattern has weighted count $O_L(p)$. For two internal pairs, excluding tags in any other set also costs $O_L(p)$. Each isolated tag removes a trace factor $1/2$, hence gives factor 2. The remaining fixed bubbles are uncrossed; shrinking one set to $p-1$ preserves the limiting crossing weight q.

Represent these prescribed bubbles by distinct unit-variance spectator variables B, C orthogonal to H in the standard q-Gaussian algebra. The known conditional expectation gives $E_H[BW_n(H)B] = q^n W_n(H)$; conditioning first onto (H, C) and then onto H yields the product of two Mehler images [1, 3]. Consequently, with $c = 4J_s^2$ and polynomial r,

$$\mathbb{E}\tau\mathcal{Q}_N(f)^\dagger\mathcal{Q}_N(g) \longrightarrow c^2 \int \overline{\mathsf{P}_q f}\, \mathsf{P}_q g \, d\nu_q, \tag{10}$$

$$\mathbb{E}\tau r(H_N)^\dagger\mathcal{Q}_N(f) \longrightarrow c \int \bar{r}\, \mathsf{P}_q f \, d\nu_q. \tag{11}$$

The second line is the earlier single-tag bridge. Hamiltonian moment convergence and expansion of the square prove (8). These spectators encode selected diagrams, not independent physical forces. Four copies of one independent probe would include additional pairings (fourth moment $2 + q$); the tag count above is essential.

3 Uniform real time without an all-degree diagram estimate

We use the previous return-expansion and charge-sum moment bounds:

$$\mathbb{E}F_i^\dagger F_i,\ \mathbb{E}F_iF_i^\dagger \le v_N 1, \qquad v_N = \frac{4J_s^2}{\lambda_N}\frac{p}{N}(1+p/N)^{2p-1}, \tag{12}$$

$$\mathbb{E}\,\mathrm{tr}_m\,\mathcal{A}_m^k \le (2k-1)!!(Nv_N)^k, \qquad \mathbb{E}\,\mathrm{tr}_m\,h^{2k} \le (2k-1)!!M^k. \tag{13}$$

M is independent of N, m: the exact Hamiltonian variance obeys $C_m \le (J_s^2/\lambda_N)e^{2p/N}$. These are Gaussian matrix moment bounds, not operator-norm bounds for individual samples. Full-Fock Schatten Hölder, including the disorder probability, gives

$$\|\mathcal{Q}_N(f)\|_2 \le \frac{2p}{N}\sum_i \|F_i\|_8^2\|f(H_N)\|_4 \tag{14}$$

$$\le 2p(7!!)^{1/4}v_N\|f(H_N)\|_4 \le C\|f(H_N)\|_4. \tag{15}$$

Independence of F_i and H_N is not used.

For real y, the integral Taylor remainder of e^{iy} has modulus at most $|y|^k/k!$. Let $T_{k-1}(s,x) = \sum_{j<k}(isx)^j/j!$. Equation (13) implies

$$\sup_{N,\,|s|\le S}\|e^{isH_N} - T_{k-1}(s,H_N)\|_4 \le \frac{S^k}{k!}\left[(4k-1)!!M^{2k}\right]^{1/4} \longrightarrow 0. \tag{16}$$

The factorial dominates the $k^{k/2}$ growth in the numerator. Apply (8) at each fixed k, then let $k\to\infty$. This order of limits avoids assuming uniform diagram errors in k.

Let $E_* = 2\sigma/\sqrt{1-q}$. To cover every finite-N spectrum, clip x to $[-E_*, E_*]$ using $\chi(x)$, and set

$$b_s(x) = c(\mathsf{P}_q e^{is\cdot})(\chi(x)), \qquad |b_s(x)| \le c. \tag{17}$$

On $[-E_*, E_*]$ this is the original conditional characteristic function; clipping supplies a bounded continuous extension. For fixed k, moment convergence and uniform integrability imply

$$\lim_N \|c(\mathsf{P}_q T_{k-1}(s,\cdot))(H_N) - b_s(H_N)\|_2^2 = c^2\|\mathsf{P}_q(T_{k-1}(s,\cdot) - e^{is\cdot})\|_{L^2(\nu_q)}^2. \tag{18}$$

The right side tends to zero since P_q is an L^2 contraction. Uniformity in $|s|\le S$ follows from the finite polynomial coefficients, (16), and a finite mesh in s; the bounded target is uniformly Lipschitz with constant at most cE_*. Combining the three approximations proves

$$\sup_{|s|\le S}\|\mathcal{Q}_N(e^{is\cdot}) - b_s(H_N)\|_2 \to 0. \tag{19}$$

At each fixed s, different bounded continuous extensions give the same limit, since the limiting spectral measure is supported on $[-E_*, E_*]$.

4 From the force sandwich to the actual ordinary kernel

Define the rectangular force dilation $D_F = KV - Vh$; it is distinct from $D = QKQ$. Exact identities are

$$D_F^\dagger D_F = \mathcal{A}_m/m, \qquad R = D_F + (p/m)Vh, \tag{20}$$

$$pD_F^\dagger e^{iKs} D_F = \frac{N}{2m}\mathcal{Q}_N(e^{is\cdot})\big|_m. \tag{21}$$

Every rectangular Schatten trace below is normalized by the same source dimension $d_m = \binom{N}{m}$. On $m \geq N/4$, (13) gives, with disorder included,

$$\|D_F\|_{4,m} = O(p^{-1/2}), \quad \|(p/m)Vh\|_{4,m} = O(p^{-1}), \quad \|\mathcal{Q}_N(e^{is\cdot})\|_{2,m} \leq C. \tag{22}$$

The last inequality also follows directly from $\|\mathcal{Q}_N(e^{is\cdot})\|_{2,m} \leq (2p/N)\|\mathcal{A}_m\|_{2,m}$, so it is uniform in *all* sectors. Expanding $R = D_F + (p/m)Vh$ and using Hölder and unitarity bounds the central-sector difference

$$\|pR^\dagger e^{iKs} R - pD_F^\dagger e^{iKs} D_F\|_{2,m} = O(p^{-1/2}), \tag{23}$$

uniformly in real s. Moreover,

$$\sum_{m \geq N/4} b_m \left|\frac{N}{2m} - 1\right|^2 = O(N^{-1}). \tag{24}$$

For the lower tail, active sectors satisfy $m \geq p$, so the same moment bounds are at worst polynomial in N, p; their binomial weight is at most $e^{-N/8}$. Inactive sectors have zero force and zero memory; the bounded target there has exponentially small norm. Thus (19)–(23) prove

$$\delta_N(S) := \sup_{|s| \leq S} \|p\Lambda(s) - b_s(H_N)\|_2 \longrightarrow 0. \tag{25}$$

This is a leading approximation, without a fluctuation-scale rate.

We also need products, so L^2 alone is insufficient. Schatten Hölder yields $\|\Lambda(s)\|_{r,m} \leq \|G\|_{r,m}$, and eigenvalue monotonicity of $0 \leq G \leq \mathcal{A}_m/m$ gives

$$\mathbb{E}\,\mathrm{tr}_m\, G^r \leq m^{-r}(2r-1)!!(Nv_N)^r \quad (r \text{ a fixed positive integer}). \tag{26}$$

The same central/tail split therefore bounds $\|p\Lambda(s)\|_8$ uniformly, and $b_s(H_N)$ is bounded. Interpolation gives

$$\sup_{|s| \leq S} \|p\Lambda(s) - b_s(H_N)\|_4 \leq C\delta_N(S)^{1/3} \longrightarrow 0. \tag{27}$$

No energy pinching, small-λ limit, or replacement of the common couplings by independent probes has been made in this step.

5 The first return is a conserved background

For $s \geq 0$, let $\Delta_s = \{(v, w) : v, w \geq 0,\ v + w \leq s\}$ and $t = s - v - w$. The exact first return is

$$J_1(s) = \int_{\Delta_s} \Lambda(v) e^{iat} \Lambda(w)\, dv\, dw. \tag{28}$$

Before approximating a or the charge prefactor, define the *exactly conserved* comparison operator on active sectors:

$$\mathcal{K}_{N,m}(s) = \frac{m}{p^2} e^{-ihs} \int_{\Delta_s} b_v(h) e^{iat} b_w(h)\, dv\, dw. \tag{29}$$

It is zero on inactive sectors. Since $m/p^2 \leq 1/\lambda_-$, the L^4 convergence (27), Hölder, and the bounded simplex area imply

$$\sup_{0 \leq s \leq S} \| m e^{-ihs/2} J_1(s) e^{-ihs/2} - \mathcal{K}_N(s) \|_2 \to 0. \tag{30}$$

For example, subtract the integrands as $(p\Lambda(v) - b_v)U(p\Lambda(w)) + b_v U(p\Lambda(w) - b_w)$. This avoids multiplying an L^2 error by an uncontrolled operator.

Combining phases in (29) leaves $e^{-i(p/m)ht}$. Dropping this factor costs at most $c^2 S^3 \|H_N\|_2/(6p)$: the prefactors cancel exactly as $(m/p^2)(p/m) = 1/p$. Charge concentration then gives $m/p^2 \to 1/(2\lambda)$ in weighted L^2; inactive tails are negligible. Thus (29) converges to the bounded function

$$\kappa_s(x) = \frac{8 J_s^4}{\lambda} \int_{\Delta_s} g_v(x) g_w(x)\, dv\, dw, \tag{31}$$

$$g_v(x) = e^{-ivx} \frac{b_v(x)}{4 J_s^2}, \qquad |\kappa_s(x)| \leq \frac{4 J_s^4}{\lambda} s^2. \tag{32}$$

For $|x| \leq E_*$, $g_v = e^{-ivx}(\mathsf{P}_q e^{iv\cdot})(x)$. Outside that interval we use (17) in (31); this keeps the phases in the proxy consistent and is bounded. For negative s set $\kappa_{-s} = \overline{\kappa_s}$. Reversing the simplex variables gives $J_1(-s) = J_1(s)^\dagger$, so all statements are uniform for $|s| \leq S$.

The prior finite-return estimate gives

$$\sup_{|s| \leq S} \| m(M(s) - \Lambda(s) - J_1(s)) \|_2 = O_S(p^{-1}). \tag{33}$$

Combining this with (30) yields the stronger unpinched statement

$$\sup_{|s| \leq S} \| m e^{-ihs/2} (M(s) - \Lambda(s)) e^{-ihs/2} - \kappa_s(H_N) \|_2 \longrightarrow 0. \tag{34}$$

On the limiting spectral interval, $\mathsf{P}_q x = qx$ also gives the check

$$\kappa_s(x) = \frac{4 J_s^4}{\lambda} s^2 + i \frac{8 J_s^4}{3\lambda} (q - 1) x s^3 + O(s^4). \tag{35}$$

The first-return contribution survives before pinching, even though its dynamic component vanishes in the limit.

6 What is reduced, and what remains to compute

The exact samplewise projection satisfies $(1-\mathsf{P}_h)\kappa_s(H_N)=0$ and is contractive in L^2. Equation (34) proves

$$\varepsilon_N(S) := \sup_{|s|\le S} \|\mathsf{F}(s) - X_0(s)\|_2 \to 0. \tag{36}$$

Thus a uniform annealed L^2 bound on the actual finite-memory field is equivalent to one on X_0. No such uniform bound is proved here.

Write $\mathcal{U}_T X = e^{iTH_N} X e^{-iTH_N}$ and define the covariance $C_N(s,u;T) = \langle \mathsf{F}(s), \mathcal{U}_T \mathsf{F}(u)\rangle$. Let $C_{0,N}$ use X_0 instead. Cauchy–Schwarz gives

$$|C_N - C_{0,N}| \le \varepsilon_N(S)\big(\|\mathsf{F}(u)\|_2 + \|X_0(s)\|_2\big), \qquad T\in\mathbb{R}. \tag{37}$$

The currently available coarse norms are $O(p)$. Hence (36) alone does *not* imply an $o(1)$ absolute covariance difference. If either field is uniformly L^2 bounded on a compact memory interval, both are, and their covariances agree asymptotically, uniformly in T. The earlier two-return, six-product reduction with $O(p^{-1})$ covariance error remains valid without that extra assumption.

The next concrete task is the subleading, common-J two-bubble fluctuation defining X_0, at two independent memory times and with exact pinching retained. Leading regression has now been upgraded to a strong result, but it cannot by itself control this p-amplified residual. Fixed-order Taylor spectra remain useful checks rather than a finite-time solution.

Saved checks. The independent script enumerates prescribed bubble pairings and constructs q-Hermite conditional polynomials using rational arithmetic. For $q = 1/4, 1/2, 3/4$, 791 five-block words per value (total Hamiltonian degree at most eight) give 2373 exact equalities. This checks the standard bubble identity, not the finite-N tag suppression proof. A separate scalar evaluation of (31) uses $J_s = 1$, $\lambda = 0.25, 0.5, 1$, energy fractions $0, 0.4, 0.85$ of E_*, and $s = 0.5, 1, 2$. Spectral orders 64/96 and simplex orders 20/32 differ by at most 1.326×10^{-13} across 27 values. This is a float64 refinement comparison, not a certified quadrature error. Code, parameters, versions and hashes accompany the local evidence record; earlier outputs are preserved.

References

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